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# Mapping user gratifications in the age of LLM-based chatbots: An affordance perspective

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ABSTRACT

Despite the widespread use of large language model (LLM)-based chatbots, little is known about what specific gratifications users obtain from the unique affordances of these systems and how these affordance-driven gratifications shape user evaluations. To address this gap, the present study maps the gratification structure of LLM chatbot use and examines whether users' primary purpose of chatbot use (information-, conversation-, or task-oriented) influences the gratifications they derive. A survey of 249 LLM chatbot users revealed nine distinct gratifications aligned with four affordance types: modality, agency, interactivity, and navigability. Purpose of use meaningfully shaped which gratifications were most salient. For example, conversational use heightened *Immersive Realism* and *Fun*, whereas information- and task-oriented use elevated *Adaptive Responsiveness*. In turn, these affordance-driven gratifications predicted key outcomes, including perceived expertise, perceived friendliness, satisfaction, attitudes, and behavioral intentions to continued use. Across outcomes, *Adaptive Responsiveness* consistently emerged as the strongest predictor, underscoring the pivotal role of contingent, high-quality dialogue in LLM-based human–AI interaction. These findings extend uses and gratifications theory and offer design implications for developing more engaging, responsive, and purpose-tailored chatbot experiences.

## 1. Introduction

Chatbots powered by artificial intelligence (AI) have become increasingly prevalent across sectors, providing services ranging from information retrieval and customer support to entertainment and personalized interaction. Among these, chatbots based on large language models (LLMs) such as GPT-3 and GPT-4 have gained prominence due to their advanced natural language understanding, context awareness, and adaptive response generation capabilities (Kathpalia, 2025). Unlike earlier rule-based or scripted chatbots, LLM-based systems leverage extensive pre-trained knowledge and fine-tuning mechanisms to generate contextually relevant, coherent, human-like responses (Jaiman, 2024; Kathpalia, 2025; Wu et al., 2023). This technical advancement enables capabilities such as retaining memory of past interactions, understanding conversational context, and interpreting emotions to provide empathetic responses, thereby offering novel interactional experiences beyond the reach of traditional chatbots (Liu, 2024). According to recent market analyses, the global chatbot market is projected to grow from $5.4 billion in 2023 to $15.5 billion by 2028, largely driven by adoption of these advanced LLM-based systems (Markets and Markets Research, 2023). Despite the rapid adoption of LLM chatbots, we still lack a clear map of how users' gratifications, central to a uses and gratifications (U&G) perspective, align with technology-specific affordances (e.g., contextual memory, adaptive response, co-creation). This study therefore addresses that gap by mapping affordance-driven gratifications of LLM-based chatbot use.

Previous scholarship has applied U&G theory to examine gratifications associated with users' engagement with AI-driven chatbots. Early studies analyzed brand-deployed commercial chatbot services (Cheng & Jiang, 2020; Marjerison et al., 2022) or chatbots used in mobile messaging platforms (Brandtzaeg & Følstad, 2018). These studies generally identified gratifications similar to those found in traditional media, such as information seeking, entertainment, social/relational interaction, novelty/curiosity, convenience, and passing time. More recent research on LLM-based chatbots, however, highlights additional gratifications such as productivity, creativity, and learning/development (Skjuve et al., 2024), as well as perceptions of intelligence and flow (Gao, 2023). Yet even these gratifications are often subsumed under the broader utilitarian, hedonic, technological, and social categories commonly applied to traditional media platforms, suggesting that

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gratifications derived from the unique affordances of LLM-based chatbots are not yet fully examined.

While LLM-based chatbots can certainly provide familiar gratifications, they also offer advanced capabilities that go beyond traditional media experiences. These include adaptive response generation (enabling personalized, context-aware replies), creative problem-solving assistance for novel or complex tasks, and affective mirroring in which chatbots recognize and respond empathetically to users' emotions (Liu, 2024). Such sophisticated functions create novel interaction experiences and user gratifications distinct from those of conventional media, indicating the need to re-examining U&G frameworks from a different perspective in order to fully capture user gratifications in the context of LLM-based chatbots.

Therefore, this study addresses that gap by investigating the distinct gratifications users derive from interacting with LLM-based chatbots, with a focus on how technological affordances shape these experiences. We review existing U&G research and then empirically examine how the affordance-driven gratifications of LLM chatbots relate to user perceptions, attitudes, satisfaction, and behavioral intentions. Ultimately, this study seeks to map the gratification landscape of LLM chatbot use, offering practical insights for designers and organizations on tailoring chatbot features to meet user expectations and enhance user engagement and satisfaction.

## 2. Literature review

### 2.1. Evolution of chatbot capabilities and implications for U&G theory

Early conversational agents were primarily rule-based chatbots designed to simulate dialogue using predefined scripts and pattern-matching algorithms (Jain et al., 2025). These systems relied on keyword recognition and deterministic decision trees, enabling only limited, domain-specific interactions (Ramanathan, 2025). As a result, early chatbots were effective for simple FAQ services or structured task support but lacked the capacity for flexible, context-aware conversation.

Recent advances in AI and natural language processing have fundamentally transformed chatbot capabilities. Unlike rule-based agents, LLM-driven chatbots are trained on vast amounts of textual data, enabling them to generate coherent, open-domain, human-like responses (Jaiman, 2024; Kathpalia, 2025; Wu et al., 2023). For example, GPT-4 can produce fluent multi-turn interactions that closely resemble human dialogue, demonstrating the context-aware capabilities of modern LLMs. Moreover, these chatbots can perform advanced cognitive and creative tasks, such as summarizing documents, drafting content, solving problems, and even providing emotional support, capabilities that were not feasible in earlier generations (Jaiman, 2024; Kathpalia, 2025; Wu et al., 2023). This technological leap has substantially expanded the interactional affordances available to users. LLM-based chatbots now provide personalization through contextually tailored responses, enhanced interactivity via fluid, real-time dialogue, multimodality by supporting text, voice, and image-based exchanges, and navigability through the ability to efficiently guide users across diverse information and topics. These affordances extend beyond the pragmatic utility of rule-based chatbots, creating opportunities for novel gratifications. Given these advancements, traditional U&G approaches, originally developed for passive media like television or radio (Katz et al., 1973), may not fully account for the motivations behind modern chatbot use. The unique capabilities of LLM chatbots suggest the need to examine U&G frameworks by focusing on the unique features of this new system.

### 2.2. Uses and gratifications of LLM-based chatbots: from needs-based to affordance-based perspectives

Uses and Gratifications theory provides a framework for understanding why people actively seek out specific media to satisfy particular needs and desires. Its central premise is that audiences are active participants who select media based on their individual needs, rather than passive recipients of content (Rubin, 2002). In high-choice environments, people choose media that best satisfy needs such as information seeking, personal identity, social integration, and entertainment (Papacharissi & Rubin, 2000; Stafford et al., 2004). Unlike traditional one-way media, online platforms allow users not only to consume but also to create content, shifting from a sender–receiver model to a dynamic, interactive paradigm (Ko et al., 2005; Van Dijk, 2000). This shift makes U&G theory especially relevant for studying engagement with interactive media (Chen, 2011).

With the emergence of AI-driven chatbots, particularly those powered by LLMs, researchers have begun applying U&G theory to explore the range of user gratifications and underlying motivations for engaging with these systems. A growing body of work shows that users derive multiple gratifications from interacting with LLM chatbots. For instance, Skjuve et al. (2024) identified six motives for ChatGPT use: productivity, learning, creative work, entertainment, novelty, and social interaction/support, demonstrating the breadth of user motivations. Similarly, Ju and Stewart (2024) found information seeking/sharing, entertainment, ease/efficiency, and social interaction as major reasons for using CahtGPT.

Context-specific studies reinforce these findings. For example, students use ChatGPT for academic content creation, information seeking, novelty, and convenience (Jishnu et al., 2023). Lee et al. (2025) identified five meaningful dimensions of ChatGPT use in educational settings: entertainment, novelty, study guide, social influence, and active interaction. Beyond academia, Zhang et al. (2025) found that families engage with ChatGPT collaboratively for information seeking, productivity, entertainment, and social bonding. Finally, qualitative analyses of user discussions on Reddit reveal gratifications of ChatGPT: utilitarian, hedonic, social gratification, and creativity enhancement (Lin & Ng, 2024).

However, despite the advanced capabilities of LLM-based chatbots, many of the gratifications identified in current literature mirror those found in earlier studies of traditional media. Classic U&G frameworks have long emphasized motives such as information seeking, hedonic enjoyment or entertainment, novelty, convenience, and social interaction. While these gratifications remain relevant, modern chatbots offer fundamentally new interaction possibilities that open the door to qualitatively distinct user experiences.

One reason empirical studies may continue to report similar gratifications is that many adopt existing typologies premised on the assumption that media use is primarily driven by pre-existing user needs. Moreover, these studies often rely on survey instruments originally developed for traditional media, which may constrain the identification of gratifications unique to LLM chatbots. While this needs-based approach has been valuable for understanding traditional media, it may not fully capture the dynamics of advanced AI systems. LLM-based chatbots offer novel affordances, defined as actionable possibilities a technology provides as perceived by users (Markus & Silver, 2008), such as adaptive response generation, creative problem-solving, affective mirroring, and real-time content personalization. These capabilities can not only fulfill existing needs in novel ways but also generate entirely new needs through use. In such cases, gratifications may emerge as a consequence of engaging with these features. Relying solely on needs-based frameworks risks overlooking these emergent, affordance-driven gratifications, highlighting the importance of perspectives that directly link specific technological capabilities to the user experiences they make possible.

Whereas traditional U&G constructs describe what users seek from media, an affordance-based interpretation explains *how* these gratifications are triggered by specific technological capabilities in real time. In other words, the former focuses on the user's pre-existing goals, whereas the latter emphasizes the system's action possibilities that may generate entirely new gratifications through use. This distinction is especially critical for LLM-based chatbots, whose adaptive and context-aware features can elicit experiences, such as immersion, realism, or dynamic tailoring, that could not be meaningfully captured within earlier needs-based categories.

Notably, the idea of incorporating media features into U&G research is not new. Lichtenstein and Rosenfeld (1983) argued that media characteristics themselves can shape gratifications, and Ruggiero (2000) emphasized that technological affordances, such as synchronicity and interactivity, create novel user experiences. Building on this affordance-centric perspective, Sundar and Limperos (2013) introduced the U&G 2.0 model, which systematically links media gratifications to four fundamental technological affordances classified by Sundar (2008): modality (the formats and sensory modes through which content is delivered), agency (the extent of user control over content or system behavior), interactivity (the degree of reciprocal, two-way communication), and navigability (the ease of exploring an information space). Each of these affordances can trigger specific mental shortcuts or heuristics that shape user enjoyment and experience. The key strength of U&G 2.0 is that it focuses on what the technology enables users to do in the moment, rather than assuming that all media use stems from fixed, pre-existing psychological needs. This makes it well-suited to capturing the novel and evolving experiences afforded by advanced AI systems.

### 2.3. Affordance-driven gratifications of LLM-based chatbots

Applied to LLM-based chatbots, technological affordances can yield a broad spectrum of user gratifications. Following Sundar and Limperos (2013), we organize these gratifications around modality, agency, interactivity, and navigability.

Modality-based gratifications typically stem from the sensory richness and format of content delivery (Sundar & Limperos, 2013). LLM-based chatbots now support multimodal content generation, producing not only text but also creative images, audio, and other sensory-rich media. This expanded capability goes beyond the traditional notion of "sensory richness," highlighting their multimodal and creative generativity. For example, systems such as GPT-4 with Vison can jointly process an image, generate an explanatory script, and produce a synthetic voiceover to deliver a short video from a single photo (OpenAI Vision Guide, 2024). It can also convert a rough hand-drawn sketch into a styled webpage mockup (Zhu et al., 2023). These cross-modal capabilities enhance immersive realism (users can "see" and "hear" the content in coordinated modalities) and enable innovative expression.

Agency-based gratifications focus on users' perceived control in the course of interaction (Sundar & Limperos, 2013). In LLM-driven chatbots, users can specify goals, set constraints, and refine outputs through step-by-step instructions or explicit preference cues, directing what the system should do and how it should respond. This ability to steer the interaction such as choosing what to perform, what to revise, and when to stop, imbues a strong sense of control or agency. Through these ongoing choices, users experience tailoring and filtering gratifications, that is, the feeling that they are actively personalizing the chatbot's outputs to fit their own needs, tastes, and priorities. Furthermore, agency extends to a social dimension. Through LLM chatbots, users can explore others' perspectives, synthesize diverse opinions, and discover emerging norms or shared knowledge, thereby building knowledge communities. This process fosters not mere conformity to popular opinions but a deeper sense of connection and bandwagon grounded in collective knowledge. Ultimately, LLM chatbots offer richer gratifications of control, personalization, and community-building.

Interactivity-based gratifications typically arise from the medium's capacity for responsive and two-way communication (Sundar & Limperos, 2013). In the case of LLM chatbots, however, they stem not simply from reciprocal exchanges but from context-aware dialogue. LLM chatbots sustain natural, flowing conversations by drawing on prior turns to respond directly to what was just said, confirming understanding, and quickly repairing misunderstandings through contextual memory and coherent turn-taking. These context-aware, dialogic features create a sense that the chatbot is truly listening and responding, which enhances feelings of reciprocity and involvement. This high quality of dialogue not only promotes user involvement but also yields gratifications such as *interaction* (feeling actively involved in a dialogue) and *activity* (the satisfaction of participating in a continuous exchange). Rather than acting as passive recipients of information, users become *co-conversants* in the dialogue.

Finally, LLM chatbots redefine how users browse and play with information. Because their interfaces are conversation-based, users can explore information simply by engaging in dialogue. Chatbots retrieve, summarize, and explain relevant information on demand, essentially acting as conversational search engines. This natural-language exploration method lets users continuously refine their queries, transforming a click-based search experience into a smooth and intuitive interaction. At the same time, LLM chatbots are inherently playful. Many interactions take a narrative or story-like form, allowing users to enjoy the process of exploration itself.

One study exemplifying this affordance-based approach is Cheng, Wang, and Lee's (2025) analysis of the World Health Organization's multilingual COVID-19 chatbot *Florence*, which was designed to combat misinformation. They identified four affordance-driven gratifications – coolness (modality), enhancement (agency), activity (interactivity), and browsing (navigability) – and found that these gratifications significantly predicted user satisfaction, engagement, and perceived relationship quality with the organization. While Cheng et al.'s findings demonstrate the value of defining gratifications in terms of what system features enable, their study examined only a subset of the broader set of affordance-based gratifications proposed by Sundar and Limperos's (2013) U&G 2.0 framework. That framework encompasses additional gratifications such as *realism*, *being there*, *coolness*, *novelty*, *filtering*, *tailoring*, *bandwagon*, *activity*, and *browsing*, which together capture a richer spectrum of experiences afforded by LLM-based chatbots.

Thus, the present study aims to empirically map a more comprehensive set of affordance-based gratifications that users derive from LLM-based chatbots. Accordingly, our first research question is posed as follows:

> **RQ1:** What gratifications do users derive from using LLM-based chatbots, particularly in relation to the four types of affordances – modality, agency, interactivity, and navigability?

### 2.4. Differences in gratifications by purpose of LLM-based chatbot use

LLM chatbots can serve a variety of user purposes, and these purposes may shape the gratifications user derive from interacting with them. Adamopoulou and Moussiades (2020) identified three major goals of chatbot use: information provision, social conversation, and task execution. Based on these goals, they outlined three corresponding chatbot types, informative, conversational, and task-based. While this categorization was well suited to earlier generations of chatbots with narrow, goal-specific functionalities, contemporary LLM-based chatbots often support multiple purposes or goals within a single interaction. As a result, it is difficult to classify an LLM chatbot as only one specific type. Nonetheless, the underlying framework remains conceptually valuable. Although LLM-based chatbots afford a wide range of functions, users typically engage with them for a primary purpose or goal in mind. These user purposes shape the usage context, which in turn activates different affordances and influences the kinds of gratifications users derive from the interaction.

For example, when users engage a chatbot with an information-oriented purpose (e.g., to obtain information or clarify a topic), they are likely to rely on navigability and modality affordances. These affordances help users access and filter information quickly and effectively. In this context, users may experience gratifications such as browsing enjoyment, a sense of realism when content is presented in vivid, concrete ways, and feelings of novelty or "coolness" when discovering new information.

When the purpose is conversational, users are more likely to draw on interactivity affordances that support responsive, back-and-forth exchanges. These affordances facilitate meaningful interaction and a sense of activity or involvement, such as feeling socially engaged during the exchange or perceiving the chatbot as responsive.

In contrast, when the purpose is task-oriented (e.g., writing code or solving math problems), users are likely to leverage agency affordances. These affordances enable users to direct the chatbot to perform actions or retrieve targeted results. The resulting gratifications include personalized filtering (receiving outputs that are tailored), in some cases, bandwagon, when users ask the chatbot to find widely endorsed options.

In sum, even when the same LLM-based chatbot is used, the primary purpose of use influences which affordances are activated and, consequently, which gratifications are most salient. Accordingly, we pose the following research question:

**RQ2:** Does the purpose of using a LLM-based chatbot (e.g., information-oriented, conversation-oriented, or task-oriented) influence the gratifications users obtain from these interactions?

### *2.5. From gratifications to attitudes, satisfaction, and behavioral intentions*

Beyond identifying gratifications, U&G theory emphasizes that they are key drivers of user outcomes such as attitudes, satisfaction, and continued usage intentions. Prior research has shown that attitudes toward a technology, defined as a positive or negative evaluation of a medium (Ajzen, 1991), are closely tied to the gratifications derived from its use (e.g., Huang et al., 2007; Ko et al., 2005; Marjerison et al., 2022). For instance, Huang et al. (2007) found that convenience and entertainment gratifications from mobile internet use were positively associated with attitudes toward mobile services. Similarly, Marjerison et al. (2022) demonstrated that utilitarian gratifications (e.g., authenticity, convenience) and hedonic gratifications (e.g., perceived enjoyment) significantly predicted favorable attitudes toward chatbots in online shopping contexts. Extending this logic to LLM chatbots, if users perceive their interactions as engaging, informative, or otherwise rewarding, they are likely to form more positive attitudes toward using such systems.

Satisfaction is another critical outcome linked to gratifications. Defined as the extent to which users' expectations and needs are fulfilled in their interaction with a technology (Chung et al., 2018), satisfaction emerges when a chatbot delivers on sought-after gratifications, for instance, by providing useful information or creating a fun, human-like conversational experience. Research indeed supports that satisfaction increases when gratifications are achieved (Cheng & Jiang, 2020; Rahi et al., 2025). Cheng, Wang, and Lee (2025) further found that affordance-based gratifications such as coolness, enhancement, activity, and browsing significantly predicted satisfaction.

Gratifications also shape future behavior. Behavioral intention, defined as the likelihood of using a medium or service in the future, is a stronger predictor of actual continued use (Oliver, 1980). Prior studies have demonstrated that higher gratification levels typically translate into stronger behavioral intentions (e.g., Alhassan et al., 2020; Chang & Zhu, 2012; Cheng & Jiang, 2020; Gan & Li, 2018). In the LLM chatbot context, if users derive substantial gratifications, whether emotional support, task efficiency, novelty, or enjoyment, they should be more inclined to continue using such systems over time. Building on these insights, we propose our third research question:

**RQ3:** Do the gratifications derived from LLM-based chatbot interactions influence users' attitudes, satisfaction, and behavioral intentions?

## 3. Method

### *3.1. Study overview*

The study examines (1) how affordance-driven gratifications emerge from users' interactions with LLM-based chatbots (RQ1), (2) how these gratifications differ by primary purpose of use (RQ2), and (3) how gratifications influence user perceptions, satisfaction, and behavioral intention (RQ3). The study overview appears in Fig. 1.

The figure summarizes the sequential analytical flow of the study. Data collection and preparation were followed by an exploratory factor analysis (EFA) to identify affordance-driven gratifications (RQ1), ANCOVA tests to examine how these gratifications differ by purposes of use (RQ2), and hierarchical regression analyses to investigate their effects on user perceptions and sustained use intention (RQ3).

### *3.2. Participants and procedure*

To answer the research questions, we conducted an online survey of LLM chatbot users. The survey was administered via CloudResearch using Prime Panels, an online platform with a large and diverse participant pool (~50 million individuals; Chandler et al., 2019). Participants were recruited based on the following criteria: they were age 18 or older, English-speaking, and had prior experience using LLM chatbots. On average, participants took about 20 min to complete the questionnaire.

A total of 249 AI chatbot users participated (136 males, 110 females, 3 identifying as other). The sample covered a broad age range: 9.6 % were 18–24 years old ($n = 24$), 24.1 % were 25–34 ($n = 60$), 18.1 % were 35–44 ($n = 45$), 19.3 % were 45–54 ($n = 48$), 16.9 % were 55–64 ($n = 42$), 10.8 % were 65–74 ($n = 27$), and 1.2 % were 75–84 ($n = 3$). In terms of ethnicity, the majority identified as Caucasian ($n = 155$), followed by African American ($n = 36$), Hispanic ($n = 27$), Asian ($n = 20$), other ethnicities ($n = 8$), with 3 participants preferring not to disclose. Regarding education, most participants held a Bachelor's degree ($n = 93$), followed by some college but no degree ($n = 58$), an Associate's degree ($n = 32$), a high school diploma or GED ($n = 27$), a Master's degree ($n = 26$), a professional degree ($n = 7$), a doctorate ($n = 4$), and less than a high school diploma ($n = 2$).

### *3.3. Measures*

#### *3.3.1. Most used chatbot*

Participants were first asked to identify the AI-powered chatbot they use most frequently. This question did not restrict responses to LLM-based chatbots. However, given the study's focus on LLM-based

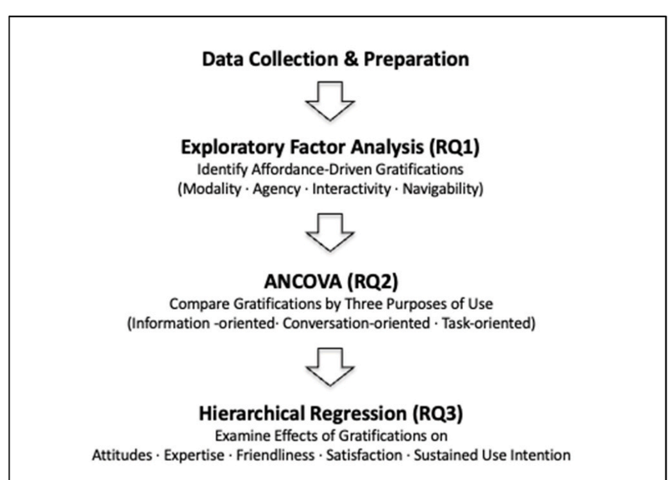


**Fig. 1.** Study overview.

chatbots, only participants who indicated using an LLM-based chatbot as their primary tool were included in the analysis. The majority (78.9 %) reported using ChatGPT, followed by Gemini (9.6 %), Copilot (8.4 %), Grok (2.4 %), and Perplexity (.8 %).

#### 3.3.2. Purpose of chatbot use

Based on the study of Adamopoulou and Moussiades (2020), we allowed participants to self-define the primary use of the chatbot: (1) Information-oriented, (2) Conversation-oriented, or (3) Task-oriented. This approach let participants define their chatbot usage based on their own experience, acknowledging that expected gratifications may vary with the purpose of chatbot use.

#### 3.3.3. Gratification obtained from chatbot use

To assess gratifications obtained, participants were instructed to think about the chatbot they use most often while responding to a series of items. Based on Sundar and Limperos's (2013) U&G 2.0 framework, we selected and adapted a set of 39 gratification statements relevant to interactions with LLM chatbots, grouped by the four affordance types (modality, agency, interactivity, and navigability). Each item began with the stem: *"I use AI chatbots because …"* and participants rated their agreement on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

*Modality-driven gratifications* included four proposed sub-dimensions: *Realism* (e.g., "The interaction feels similar to talking to a real person"; $M = 4.29$, $SD = 1.32$; Cronbach's $\alpha = .78$), *Coolness* (e.g., "They provide a unique user experience"; $M = 4.44$, $SD = 1.29$; $\alpha = .82$), *Novelty* (e.g., "They offer a new way of interacting with technology"; $M = 5.47$, $SD = .97$; $\alpha = .82$), and *Being There* (e.g., "They help me immerse myself in places that I cannot physically experience"; $M = 3.42$, $SD = 1.60$; $\alpha = .94$).

*Agency-driven gratifications* included *Community-Building* (e.g., "They help expand my social interactions"; $M = 3.06$, $SD = 1.65$; $\alpha = .96$), *Bandwagon* (e.g., "They allow me to see what others think before I make a decision"; $M = 3.89$, $SD = 1.66$; $\alpha = .93$), and *Filtering* (e.g., "They help me filter through information"; $M = 4.77$, $SD = 1.31$; $\alpha = .78$). We did not include items for "*agency enhancement*" as defined in some social media contexts (e.g., identity expression or broadcasting to others), since interacting with chatbots typically does not involve those social broadcasting functions.

*Interactivity-driven gratifications* included *Interaction* (e.g., "I expect to have ongoing interactions with them"; $M = 5.37$, $SD = 1.13$; $\alpha = .78$), *Activity* (e.g., "I feel active when using them"; $M = 4.94$, $SD = 1.24$; $\alpha = .73$), and *Responsiveness* (e.g., "They are responsive to my commands and needs"; $M = 5.47$, $SD = 1.03$; $\alpha = .81$). We excluded items related to *Dynamic Control*, which refers to a user's fine-grained control over the system's internal behavior. In the chatbot context, users do influence the interaction through their queries, but they typically do not control the system's inner workings in a step-by-step, adjustable manner. Thus, dynamic control was not considered a core aspect of typical chatbot use and those items were omitted.

*Navigability-driven gratifications* included *Browsing* (e.g., "They allow me to access a wide variety of information quickly"; $M = 5.85$, $SD = 1.00$; $\alpha = .87$) and *Fun* (e.g., "They are fun to explore and interact with"; $M = 5.13$, $SD = 1.35$; $\alpha = .89$). We did not include a *Scaffolding* gratification (related to guided, stepwise learning or interaction), because chatbot interactions are generally free-form and user-driven rather than following a predetermined guided sequence.

#### 3.3.4. Attitudes toward chatbots

General attitudes toward AI chatbots were measured using ten 7-point semantic differential items adapted from Sundar et al. (2011). Participants rated the extent to which they found chatbots, for example, *unappealing–appealing, useless–useful, negative–positive, bad–good, unattractive–attractive, boring–exciting, unpleasant–pleasant, unlikable–likable, uninteresting–interesting*, and *low-quality–high-quality*. These items were averaged to form an overall attitude scale ($M = 5.37$, $SD = 1.01$; $\alpha = .94$). In addition, two specific perception scales from prior research (Koh & Sundar, 2010) were included: Perceived Expertise of the chatbot (four items assessing whether the chatbot is *intelligent, knowledgeable, competent*, and *informed*; $M = 5.37$, $SD = 1.01$; $\alpha = .94$) and Perceived Friendliness of the chatbot (seven items assessing whether the chatbot is *empathetic, personal, warm, emotionally invested, willing to listen, careful*, and *open*; $M = 4.03$, $SD = 1.52$; $\alpha = .94$).

#### 3.3.5. Satisfaction with chatbot interaction

Satisfaction with the chatbot was measured with a single item: *"Overall, how satisfied are you with your interactions with the chatbot you use most frequently?"* Participants responded on a 7-point scale (1 = extremely dissatisfied, 7 = extremely satisfied). The mean satisfaction rating was 5.62 ($SD = 1.40$).

#### 3.3.6. Behavioral intention

Participants' intention to continue using chatbots was assessed with two Likert-type items adapted from De Cosmo et al. (2021): *"I intend to use chatbots in the near future"* and *"I believe my interest in chatbots will increase in the future."* These were rated on 7-point agreement scales and averaged ($M = 5.91$, $SD = 1.12$; $\alpha = .83$).

## 4. Results

To provide an overview of the key variables, Table 1 reports descriptive statistics (means and standard deviations), interquartile ranges, and Cronbach's alpha reliability coefficients for each measured construct. As shown, all multi-item variables demonstrated acceptable internal consistency ($\alpha > .70$).

**Table 1**
Descriptive statistics and reliability coefficients for key variables.

| Variable | *M* | *SD* | 25th Percentile (Q1) | 75th Percentile (Q3) | Cronbach's *α* |
|---|---|---|---|---|---|
| Realism | 4.29 | 1.32 | 3.33 | 5.33 | .78 |
| Coolness | 4.44 | 1.29 | 4.00 | 5.33 | .82 |
| Novelty | 5.47 | .97 | 5.00 | 6.00 | .82 |
| Being There | 3.42 | 1.60 | 2.00 | 4.67 | .94 |
| Community-Building | 3.06 | 1.65 | 1.88 | 4.50 | .96 |
| Bandwagon | 3.89 | 1.66 | 2.67 | 5.00 | .93 |
| Filtering | 4.77 | 1.31 | 4.00 | 5.67 | .78 |
| Interaction | 5.37 | 1.13 | 4.67 | 6.33 | .78 |
| Activity | 4.94 | 1.24 | 4.33 | 6.00 | .73 |
| Responsiveness | 5.47 | 1.03 | 5.00 | 6.00 | .80 |
| Browsing | 5.85 | 1.00 | 5.50 | 6.50 | .87 |
| Fun | 5.13 | 1.35 | 4.33 | 6.00 | .89 |
| Attitudes | 5.37 | 1.01 | 4.80 | 6.00 | .94 |
| Perceived Expertise | 5.56 | 1.31 | 5.00 | 6.50 | .94 |
| Perceived Friendliness | 4.03 | 1.52 | 3.00 | 5.14 | .94 |
| Satisfaction | 5.62 | 1.40 | 5.00 | 6.00 | N/A |
| Behavioral Intention | 5.91 | 1.12 | 5.50 | 7.00 | .83 |
| Immersive Realism | 3.66 | 1.44 | 2.60 | 4.80 | .91 |
| Innovation | 5.35 | 1.03 | 4.80 | 6.00 | .87 |
| Personalized Filtering | 4.70 | 1.27 | 4.10 | 5.60 | .85 |
| Adaptive Responsiveness | 5.63 | .95 | 5.00 | 6.25 | .83 |
| Participatory Interaction | 4.70 | 1.44 | 4.00 | 6.00 | .76 |

Note. All factors demonstrated acceptable reliability, with Cronbach's $\alpha$ ranging from .73 to .96. According to conventional benchmarks (Nunnally & Bernstein, 1994), these values indicate good to excellent internal consistency.

### 4.1. Gratifications derived from LLM-based chatbot use (RQ1)

We first examined whether the gratification items mapped onto the four theorized affordance groups (modality, agency, interactivity, and navigability). A series of exploratory factor analyses (EFAs) were conducted on all 39 gratification items. Because the study aimed to identify latent constructs rather than maximize explained variance, we employed principal axis factoring (PAF) rather than principal components extraction (PCA). Given that the latent constructs were expected to correlate conceptually, an oblique (Promax) rotation was applied. Sampling adequacy was excellent (KMO = .917), and Bartlett's test of sphericity was significant ($\chi^2(741) = 7992.84$, $p < .001$). The initial EFA yielded a clear four-factor structure consistent with the theorized four types of affordances. Items loaded strongly (.60-.86) on their primary factors with minimal cross-loadings (<.35). The four-factor solution accounted for approximately 55.94 % of the total variance and demonstrated good internal consistency (α ≥ .74).

For additional robustness checks, several supplementary analyses were conducted. First, a rotation-sensitivity test was performed by re-estimating the four-factor solution using orthogonal (Varimax) rotation while fixing the number of factors to four. The resulting structure remained stable, with strong primary loadings (.60-.86), minimal cross-loadings (<.35), and a cumulative variance of 54.49 % explained (KMO = .917; Bartlett's $\chi^2(741) = 7992.84$, $p < .001$). This finding demonstrates that the factor pattern is not dependent on a particular rotation method. Second, a split-sample EFA was conducted to examine factor stability across subsamples. The dataset was randomly divided into two halves. Both subsamples yielded consistent four-factor solutions (KMO = .856 and .885; Bartlett's $\chi^2(741) = 4578.78$ and 3950.19, respectively), explaining 56.66 % and 59.33 % of total variance. Overall, the analyses supported a four-affordance taxonomy, modality, agency, interactivity, and navigability, each represented by theoretically and empirically separable factors.

In addition to the overall EFAs on all 39 gratification items, we conducted supplementary EFAs separately for each affordance domain – modality, agency, interactivity, and navigability. Given that each affordance type represents a conceptually distinct set of gratifications, separate analyses provided a more fine-grained assessment of how items cohered within and across affordance categories. The following section presents the results of individual EFAs conducted for each affordance domain.

### 4.2. Modality-driven gratifications

The EFA for modality-related items initially led to the removal of three items due to low communalities or ambiguous content (“I know the information they provide is real” from the *Realism* set, “They are stylish” from *Coolness*, and “They offer an unusual experience” from *Novelty*). After removing these items, the remaining modality items yielded two factors, which together explained 64.62 % of the variance (Bartlett's $\chi^2(45) = 2073.31$, $p < .001$; KMO = .816). Factor 1 can be labeled *Immersive Realism* and Factor 2 labeled *Innovation* (see Table 2 for factor loadings). Factor 1 had an eigenvalue of 5.32 and Factor 2 an eigenvalue of 1.81.

Factor 1, *Immersive Realism*, reflects users' perception of the chatbot's ability to simulate authentic, engaging, lifelike interactions that create a feeling of being present in non-physical or distant environments. High-loading items described feeling involved in far-away or imagined environments and interactions that mirror real-life conversations. This factor highlights the chatbot's role in making users feel immersed, involved, and *“present”* in a realistic way.

Factor 2, *Innovation*, captures users’ perception of the chatbot as a groundbreaking, novel, and distinctive technology. The items loading on this factor emphasize that the chatbot provides a new way of interacting with technology and is perceived as an innovative tool or experience.

**Table 2**
Factor analysis on perceived modality-driven gratifications of LLM-based chatbots.

| $N = 249$ | Factor Loading | |
|---|---|---|
| I use AI chatbots because … | 1 | 2 |
| ***1. Immersive Realism*** | | |
| They create a sense of being involved in distant environments. | **.993** | −.134 |
| They help me immerse myself in places that I cannot physically experience. | **.942** | −.106 |
| They allow me to experience things without actually being there. | **.883** | −.055 |
| The interaction feels similar to talking to a real person. | **.564** | .275 |
| The experience closely mirrors real-life interactions. | **.550** | .272 |
| ***2. Innovation*** | | |
| They offer a new way of interacting with technology. | −.091 | **.855** |
| The technology is innovative. | −.047 | **.831** |
| They are different from traditional tools. | .172 | **.710** |
| They provide a distinctive user experience. | −.149 | **.681** |
| They provide a unique user experience. | .188 | **.679** |
| Variance Explained (%) | 49.80 | 14.82 |
| Eigenvalue | 5.32 | 1.81 |
| Mean Score | 3.66 | 5.35 |
| *SD* | 1.44 | 1.03 |

### 4.3. Agency-driven gratifications

EFA on the agency-related items produced three factors, explaining 71.60 % of the variance ($\chi^2(66) = 2586.28$, $p < .001$; KMO = .896). One item from an “*Ownness*” category (“The content reflects my true self, mirroring my identity or values”) was dropped due to cross-loading on multiple factors. **Factor 1** was defined as *Community-Building*, Factor 2 as *Personalized Filtering*, and Factor 3 as *Bandwagon*. Table 3 presents the factor loadings.

**Table 3**
Factor analysis on perceived agency-driven gratifications of LLM-based chatbots.

| $N = 249$ | Factor Loading | | |
|---|---|---|---|
| I use AI chatbots because … | 1 | 2 | 3 |
| ***1. Community-Building*** | | | |
| They help me connect with others. | **.989** | .026 | −.074 |
| They help expand my social interactions. | **.942** | −.037 | .007 |
| They allow me to build relationships or interactions that feel valuable. | **.858** | .011 | .085 |
| They make me feel like I am part of a larger user community. | **.843** | −.021 | .089 |
| ***2. Personalized Filtering*** | | | |
| They help me filter through information. | −.190 | **.690** | .057 |
| I can avoid seeing content or information that I don't want. | .147 | **.680** | −.112 |
| They let me set my preferences. | .079 | **.723** | −.002 |
| Their customizable interaction makes the experience feel uniquely mine. | −.002 | **.740** | .102 |
| They offer a personalized experience that makes me feel like it's tailored just for me. | .038 | **.725** | −.004 |
| ***3. Bandwagon*** | | | |
| They give me comfort by showing the opinions of others. | .025 | −.009 | **.900** |
| They allow me to see what others think before I make a decision. | .015 | .031 | **.855** |
| They allow me to compare my thoughts with those of others. | .045 | .013 | **.886** |
| Variance Explained (%) | 53.88 | 9.96 | 7.76 |
| Eigenvalue | 6.71 | 1.55 | 1.11 |
| Mean Score | 3.06 | 4.70 | 3.89 |
| *SD* | 1.65 | 1.27 | 1.66 |

Factor 1, *Community-Building*, encompassed items indicating that the chatbot helps the user connect with others, expand social interactions, and feel part of a larger community. It represents social/relational gratifications. Users feel the chatbot enables meaningful interactions or a sense of belonging.

Factor 2, *Personalized Filtering*, combined items originally under *Filtering* with a couple of items from the dropped *Ownness* category. It represents the chatbot's ability to customize or tailor content to the user. Users value that the chatbot can filter information, avoid unwanted content, and allow them to set preferences to meet their specific needs. Essentially, it reflects gratification from personalized content curation by the chatbot.

Factor 3, *Bandwagon*, relates to gratifications obtained from seeing others' opinions or popular trends via the chatbot. Items suggest that users use the chatbot as a way to gauge what others think by comparing their own thoughts with those of others or getting social reassurance before making decisions.

### 4.4. Interactivity-driven gratifications

The EFA for interactivity items resulted in two factors, explaining 58.26 % of the variance ($\chi^2(15) = 587.29$, $p < .001$; KMO = .786). We removed three items that cross-loaded on multiple factors ("I expect to have ongoing interactions …" from the *Interaction* subscale, "I can do many things or tasks within one platform" from *Activity*, and "They can anticipate my needs based on previous interactions" from *Responsiveness*). The two retained factors were *Adaptive Responsiveness* (Factor 1) and *Participatory Interaction* (Factor 2). Factor eigenvalues were 3.27 and 1.04, respectively (Table 4).

Factor 1, *Adaptive Responsiveness,* reflects the gratification users get from the chatbot's ability to dynamically adjust to their inputs and needs. High-loading items indicated that the chatbot responds well to user requests, caters to their preferences, and helps accomplish a wide variety of tasks. It emphasizes the chatbot's flexibility and *responsiveness* in meeting individual user demands.

Factor 2, *Participatory Interaction*, represents the extent to which the interaction feels engaging and requires active user involvement. Items suggest that using the chatbot feels like an active, not passive, experience. Users feel they are actively participating in the exchange. The gratification here is derived from the interactive, two-way nature of the experience that keeps the user actively engaged.

**Table 4**
Factor analysis on perceived interactivity-driven gratifications of LLM-based chatbots.

| *N* = 249 | Factor Loading | |
|---|---|---|
| I use AI chatbots because … | 1 | 2 |
| ***1. Adaptive Responsiveness*** | | |
| They let me continuously specify and adjust my needs and preferences. | **.858** | −.102 |
| They allow me to accomplish a wide variety of tasks. | **.796** | −.071 |
| They are responsive to my commands and needs. | **.649** | .177 |
| They respond well to my requests and inputs. | **.584** | .170 |
| ***2. Participatory Interaction*** | | |
| The interaction is not passive but involves active participation. | −.039 | **.823** |
| I feel active when using them. | .024 | **.737** |
| Variance Explained (%) | 47.48 | 10.79 |
| Eigenvalue | 3.27 | 1.04 |
| Mean Score | 5.63 | 4.70 |
| *SD* | .95 | 1.44 |

### 4.5. Navigability-driven gratifications

The EFA for navigability items yielded two factors, accounting for 68.23 % of the variance ($\chi^2(21) = 1025.78$, $p < .001$; KMO = .804). We labeled these factors *Browsing* (Factor 1) and ***Fun*** (Factor 2) (Table 5).

Factor 1, *Browsing*, included items about searching, exploring, and skimming through a breadth of content using the chatbot. It relates to the gratification of being able to quickly access a wide variety of information and freely browse content via the chatbot.

Factor 2, *Fun*, encompassed items reflecting entertainment and enjoyment in using the chatbot. It captures the *fun* or playful aspect of interacting with the chatbot (e.g., using the chatbot as an enjoyable escape from routine tasks or engaging in playful interactions).

In total, our analyses identified nine distinct gratification factors across the four affordance categories: Modality (*Immersive Realism and Innovation), Agency (Community-Building, Personalized Filtering, and Bandwagon), Interactivity (Adaptive Responsiveness and Participatory Interaction), and Navigability (Browsing and Fun)*. These factors are used in subsequent analyses.

### 4.6. Differences by primary purpose of chatbot use (RQ2)

We next examined whether different purposes of chatbot use lead to different levels of these gratifications and other user outcomes (RQ2). A series of ANCOVA tests were conducted, treating the purpose of chatbot use (information-oriented vs. conversation-oriented vs. task-oriented) as the independent variable, while controlling for demographic variables (gender, age, education, ethnicity) as covariates. Table 6 summarizes the results.

Among the nine gratification factors, *Immersive Realism* (Modality gratification)*, Adaptive Responsiveness* (Interactivity gratification), and *Fun* (Navigability gratification) varied by primary purpose of LLM chatbot use. *Immersive Realism* differed significantly across groups, $F(2, 242) = 5.38$, $p < .01$, partial $\eta^2 = .043$. Post-hoc comparisons indicated that users who primarily use chatbots for conversation ($M = 4.10$) or information ($M = 3.67$) reported higher *Immersive Realism* than those using them for task completion ($M = 3.12$, $p < .05$), with no significant difference between conversation and information ($p = .068$). *Adaptive Responsiveness* also differed by purpose, $F(2, 242) = 3.61$, $p < .05$, partial

**Table 5**
Factor analysis on perceived navigability-driven gratifications of LLM-based chatbots.

| *N* = 285 | Factor Loading | |
|---|---|---|
| I use AI chatbots because … | 1 | 2 |
| ***1. Browsing*** | | |
| They let me search for things I'm interested in. | **.881** | −.022 |
| They allow me to access a wide variety of information quickly. | **.796** | −.048 |
| They allow me to browse freely through content or information. | **.823** | .005 |
| They help me skim through content and links efficiently. | **.656** | .115 |
| ***2. Fun*** | | |
| They provide an enjoyable escape from routine tasks. | −.037 | **.924** |
| They allow me to engage in playful or enjoyable interactions. | −.026 | **.913** |
| They are fun to explore and interact with. | .096 | **.718** |
| Variance Explained (%) | 49.51 | 18.72 |
| Eigenvalue | 3.78 | 1.59 |
| Mean Score | 5.85 | 5.13 |
| *SD* | 1.00 | 1.35 |

**Table 6**
Results of ANCOVA testing the effects of purpose of chatbot use on gratifications, user perceptions, and behavioral outcomes.

| Dependent Variable | $F$ | $p$ | Partial $\eta^2$ (Effect Size) |
|---|---|---|---|
| ***Modality Gratification*** | | | |
| Immersive Realism | 5.377 | **< .01** | .043 |
| Innovation | .618 | .540 | .005 |
| ***Agency Gratification*** | | | |
| Community-Building | 1.145 | .320 | .009 |
| Personalized Filtering | 2.136 | .120 | .017 |
| Bandwagon | 1.520 | .221 | .012 |
| ***Interactivity Gratification*** | | | |
| Adaptive Responsiveness | 3.610 | **< .05** | .029 |
| Participatory Interaction | 1.634 | .197 | .013 |
| ***Navigability Gratification*** | | | |
| Browsing | 2.864 | .059 | .023 |
| Fun | 3.153 | **< .05** | .025 |
| Attitudes | 1.421 | .243 | .012 |
| Perceived Expertise | 3.577 | **< .05** | .029 |
| Perceived Friendliness | 4.385 | **< .05** | .035 |
| Satisfaction | 2.758 | .065 | .022 |
| Behavioral Intention | 2.679 | .071 | .022 |

$\eta^2 = .029$. Users who use LLM chatbots for information ($M = 5.75$) and task completion ($M = 5.50$) scored higher than those using them for conversation ($M = 5.36$, $p < .05$), with no difference between information and task completion ($p = .068$). Finally, the *Fun* gratification differed by purpose of use, $F(2, 242) = 3.15$, $p < .05$, $\eta^2 = .025$. Post-hoc tests showed that using LLM chatbots for conversation (M = 5.50) yielded higher *Fun* than using them for task completion (M = 4.81; $p < .05$), with no difference between information (M = 5.10) and task completion ($p = .201$), and between information and conversation ($p = .072$).

We also examined whether the purpose of chatbot use influenced participants' attitudes toward chatbots, perceived chatbot qualities (expertise and friendliness), satisfaction, and behavioral intentions. General attitude toward chatbots did not significantly differ by the purpose, $F(2, 242) = 1.42$, $p = .243$, suggesting that whether one primarily for information, conversation, or task completion does not, in itself, lead to more positive or negative overall attitudes. However, perceived expertise of the chatbot was significantly affected by the purpose, $F(2, 242) = 3.58$, $p < .05$, $\eta^2 = .029$. Specifically, those using LLM chatbots for information ($M = 5.71$) and conversation ($M = 5.41$) rated the chatbots as more expert/competent than those using them for task completion ($M = 5.15$, $p < .05$), with no difference between information and conversation ($p = .160$). Perceived friendliness of the chatbot also differed by the purpose, $F(2, 242) = 4.39$, $p < .05$, $\eta^2 = .035$. Users using LLM chatbots for conversation ($M = 4.48$, $p < .01$) perceived the chatbot as friendlier/warmer than those using chatbots for task completion ($M = 3.54$), with no difference between information ($M = 4.02$) and conversation ($p = .069$), and between task completion and information ($p = .066$).

Satisfaction with LLM chatbots did not significantly differ by the purpose, $F(2, 242) = 2.76$, $p = .065$. Finally, behavioral intention to continue using chatbots did not significantly differ by the purpose, $F(2, 242) = 2.68$, $p = .071$.

### *4.7. Effects of gratifications on attitudes, satisfaction, and behavioral intention (RQ3)*

Our third research question asked whether the gratifications influence key outcome variables. We ran a series of hierarchical linear regressions for each outcome (attitudes, perceived expertise, perceived friendliness, satisfaction, and behavioral intention). In Step 1 of each regression, we entered the demographic controls (gender, age, education, ethnicity), and in Step 2 we entered the nine gratification factor scores as predictors. For brevity, we report here the results of the full models (including gratifications); demographic variables alone did not explain significant variance in any outcome.

**Table 7**
Summary of hierarchical regression models predicting key outcomes.

| Dependent Variable | $R$ | $R^2$ | Adjusted $R^2$ | $F$ | $p$ |
|---|---|---|---|---|---|
| Attitudes | .830 | .689 | .671 | 39.933 | <.001 |
| Perceived Expertise | .693 | .480 | .451 | 16.695 | <.001 |
| **Perceived Friendliness** | .703 | .495 | .467 | 17.701 | <.001 |
| Satisfaction | .511 | .261 | .220 | 6.375 | <.001 |
| Behavioral Intention | .709 | .503 | .476 | 18.312 | <.001 |

For attitudes toward chatbots, the model including gratifications was significant, $F(13, 235) = 39.993$, $p < .001$, and explained 68.9 % of the variance in general attitude (adjusted $R^2 = .671$). Significant positive predictors of more favorable attitudes were *Immersive Realism* ($\beta = .129$, $p < .01$), *Innovation* ($\beta = .111$, $p < .05$), *Adaptive Responsiveness* ($\beta = .324$, $p < .001$), *Participatory Interaction* ($\beta = .072$, $p < .05$), *Browsing* ($\beta = .120$, $p < .05$), and *Fun* ($\beta = .212$, $p < .001$). These showed that modality, interactivity, and navigability gratifications contributed to attitudes toward chatbots, but not agency gratification.

For perceived expertise of the chatbot, the full model was significant, $F(13, 235) = 16.695$, $p < .001$, with $R^2 = .480$ (adj. = .451). Three gratifications emerged as significant positive predictors of perceiving the chatbot as more expert-like or competent: *Immersive Realism* ($\beta = .214$, $p < .01$), *Adaptive Responsiveness* ($\beta = .375$, $p < .001$) and *Browsing* ($\beta = .284$, $p < .01$). Each of from modality, interactivity, and navigability gratifications.

For perceived friendliness of the chatbot, the model was significant, $F(13, 235) = 17.701$, $p < .001$, with $R^2 = .495$ (adj. = .467). Gratifications that significantly predicted higher friendliness perceptions were *Immersive Realism* ($\beta = .195$, $p < .05$), *Personalized Filtering* ($\beta = .268$, $p < .01$), *Participatory Interaction* ($\beta = .140$, $p < .05$), and *Fun* ($\beta = .233$, $p < .01$), each of from modality, agency, and navigability gratification.

For satisfaction with the chatbot, the regression was significant, $F(13, 235) = 6.375$, $p < .001$, with $R^2 = .261$ (adj. = .220). Notably, the *only* significant predictor of overall satisfaction was *Adaptive Responsiveness* ($\beta = .448$, $p < .001$). This indicates that, among all gratification types, the feeling that the chatbot responds adaptively to one's needs was the strongest driver of overall satisfaction.

For behavioral intention to continue using chatbots, the model was significant, $F(13, 235) = 18.312$, $p < .001$, with $R^2 = .503$ (adj. = .476). Three gratifications were significant positive predictors of intention to use chatbots in the future: *Adaptive Responsiveness* ($\beta = .381$, $p < .001$), *Browsing* ($\beta = .254$, $p < .001$), and *Fun* ($\beta = .117$, $p < .05$), one from interactivity, and two from navigability (Tables 7 and 8).

## 5. Discussion

### *5.1. Key findings*

#### *5.1.1. Key affordances and user gratifications*

This study examined user gratifications in the context of LLM-based chatbots by focusing on the technological affordances. By shifting the analytical lens toward affordances, the "action possibilities" enabled by LLM chatbots, we gain deeper insight into why users find these systems gratifying, beyond what traditional needs-based frameworks might explain.

Our findings identified nine distinct gratifications mapped onto four key affordance types. Modality affordances, reflecting the representational richness of LLM outputs, give rise to two key gratifications. *Immersive Realism* captures users' feelings that interactions are authentic, life-like, and create a sense of "being there." This aligns with the concept

**Table 8**
Regression coefficients for predictors of key outcomes.

| Dependent Variable | | Predictor | B | SE B | β | t | p |
|---|---|---|---|---|---|---|---|
| Attitudes | Modality | Immersive Realism | .129 | .042 | .185 | 3.062 | **< .01** |
| | | Innovation | .111 | .054 | .113 | 2.052 | **< .05** |
| | Agency | Community-Building | −.014 | .036 | −.023 | −.403 | .687 |
| | | Personalized Filtering | .038 | .045 | .047 | .841 | .401 |
| | | Bandwagon | −.027 | .031 | −.044 | −.859 | .391 |
| | Interactivity | Adaptive Responsiveness | .324 | .054 | .306 | 5.995 | **< .001** |
| | | Participatory Interaction | .072 | .036 | .103 | 2.024 | **< .05** |
| | Navigability | Browsing | .120 | .051 | .119 | 2.351 | **< .05** |
| | | Fun | .212 | .039 | .283 | 5.447 | **< .001** |
| Perceived Expertise | Modality | Immersive Realism | .214 | .071 | .235 | 3.010 | **<.01** |
| | | Innovation | .039 | .091 | .031 | .431 | .667 |
| | Agency | Community-Building | −.057 | .060 | −.072 | −.957 | .340 |
| | | Personalized Filtering | .115 | .075 | .112 | 1.534 | .126 |
| | | Bandwagon | .005 | .053 | .006 | .096 | .924 |
| | Interactivity | Adaptive Responsiveness | .375 | .091 | .272 | 4.120 | **< .001** |
| | | Participatory Interaction | .106 | .060 | .115 | 1.758 | .080 |
| | Navigability | Browsing | .284 | .086 | .216 | 3.313 | **< .001** |
| | | Fun | −.010 | .065 | −.010 | −.152 | .879 |
| **Perceived Friendliness** | Modality | Immersive Realism | .195 | .081 | .185 | 2.408 | **< .05** |
| | | Innovation | −.152 | .104 | −.103 | −1.470 | .143 |
| | Agency | Community-Building | .080 | .068 | .087 | 1.167 | .245 |
| | | Personalized Filtering | .268 | .086 | .224 | 3.126 | **< .01** |
| | | Bandwagon | .067 | .060 | .073 | 1.108 | .269 |
| | Interactivity | Adaptive Responsiveness | .203 | .104 | .127 | 1.953 | .052 |
| | | Participatory Interaction | .140 | .069 | .132 | 2.047 | **<. 05** |
| | Navigability | Browsing | −.074 | .098 | −.049 | −.761 | .447 |
| | | Fun | .233 | .075 | .207 | 3.120 | **< .01** |
| Satisfaction | Modality | Immersive Realism | .138 | .091 | .142 | 1.521 | .130 |
| | | Innovation | −.007 | .116 | −.005 | −.064 | .949 |
| | Agency | Community-Building | .011 | .076 | .014 | .151 | .880 |
| | | Personalized Filtering | .122 | .077 | .125 | 1.593 | .113 |
| | | Bandwagon | .030 | .067 | .036 | .448 | .655 |
| | Interactivity | Adaptive Responsiveness | .448 | .116 | .304 | 3.861 | **< .001** |
| | | Participatory Interaction | .122 | .077 | .125 | 1.593 | .113 |
| | Navigability | Browsing | .075 | .109 | .053 | .682 | .496 |
| | | Fun | −.012 | .083 | −.011 | −.139 | .889 |
| Behavioral Intention | Modality | Immersive Realism | .107 | .060 | .137 | 1.795 | .074 |
| | | Innovation | .101 | .076 | .092 | 1.321 | .188 |
| | Agency | Community-Building | −.022 | .050 | −.032 | −.436 | .663 |
| | | Personalized Filtering | .059 | .050 | .075 | 1.162 | .247 |
| | | Bandwagon | −.016 | .044 | −.023 | −.353 | .725 |
| | Interactivity | Adaptive Responsiveness | .363 | .076 | .307 | 4.751 | **< .001** |
| | | Participatory Interaction | .059 | .050 | .075 | 1.162 | .247 |
| | Navigability | Browsing | .254 | .072 | .225 | 3.527 | **< .001** |
| | | Fun | .117 | .055 | .139 | 2.123 | **< .05** |

of presence in media experience (Lombard & Ditton, 1997), which enables users to feel transported and socially present with the chatbots. The second, *Innovation*, reflects users' perceptions of novelty and the "coolness" factor of LLM chatbots. Agency affordances, rooted in personalization and source-related capabilities, pertain to who or what shape the content. Three gratifications emerged in this category. *Personalized Filtering* reflects users' recognition that responses are tailored to their individual needs and preferences. *Bandwagon* gratification arises when LLM chatbots leverage the crowd's opinions, reviews, or popular viewpoints, providing users with social proof and reassurance. This corresponds to the "bandwagon heuristic" in online information that users take the endorsement of many others as a cue to credibility (Sundar, 2008). Finally, *Community-Building* gratification captures the social and relational value users derive from their interactions with LLM chatbots. Even though users are technically communicating with an AI, they often experience a sense of social connection or community, a pattern consistent with earlier findings that people can experience interpersonal utility and companionship through media use (Papacharissi & Rubin, 2000). Interactivity affordances, enabled by the dialogic and responsive nature of LLM chatbots, enhance gratifications centered on user engagement and co-creation. *Adaptive Responsiveness* refers to the chatbot's ability to adjust in real time to user input, remember context across turns, and provide contextually coherent responses, creating smooth conversational flow. *Participatory Interaction* represents the perception that conversations are co-created, dynamic exchanges in which users actively shape the interaction rather than passively receiving information. Finally, navigability affordances, linked to the chatbot's ability to support open-ended information exploration, generate two gratifications. *Browsing* reflects the freedom to move fluidly across topics in a conversational search environment, while *Fun* captures the enjoyment that emerges from playful, curiosity-driven exploration with the chatbot. Taken together, these nine gratifications offer a comprehensive understanding of why users find LLM-based chatbots compelling. Notably, these findings extend traditional U&G categories by incorporating affordance-driven gratifications that emerge directly from what these AI systems make possible in human-AI interaction.

#### 5.1.2. Differences by use purpose in gratifications

Importantly, our findings further indicate that users' primary purpose for using LLM chatbots meaningfully shapes the types of gratifications they derive. This is consistent with the U&G perspective that different usage contexts lead to different gratification sought and obtained (Rubin, 1983). Three gratifications varied notably by use purpose: *Immersive Realism*, *Adaptive Responsiveness*, and *Fun*. Users who mainly engaged in conversation or information seeking reported greater *Immersive Realism* than those who used chatbots primarily for task completion. This suggests that exploratory or dialogic use contexts are more likely to elicit feelings of being "drawn into" the interaction and perceiving it as more life-like. In contrast, task-oriented users, focused on utilitarian goals, tended to approach LLM chatbots more instrumentally, reducing the likelihood of feeling "transported" or emotionally engaged. These findings resonate with prior research on interactive web experiences that more exploratory use of media is associated with deeper engagement and presence (Novak et al., 2000).

*Adaptive Responsiveness* was rated higher among users who used chatbots for information seeking or task completion than those using them for conversation. This pattern suggests that, in goal-oriented contexts, users appear to be especially sensitive to the chatbot's ability to understand their intent, adjust its responses, and provide clear, context-appropriate answers, as these aspects of adaptive responsiveness directly affect goal completion. In contrast, in purely conversational use, users might be less concerned with goal completion, and instead focus more on enjoying the interaction itself. This finding underscores that in practical or utilitarian contexts, the interactive quality and adaptiveness of LLM chatbots are critical. This aligns with HCI research showing that, in customer service settings where chatbots are deployed to solve users' problems, their responsiveness is a key driver of positive user experiences (Go & Sundar, 2019; Chen et al., 2021).

Conversely, *Fun* was more strongly associated with conversational rather than task-oriented use. When users approach the chatbot as a conversational partner, the interaction itself becomes the reward, eliciting delight and curiosity. In contrast, task-focused use is typically experienced in a more utilitarian, efficiency-driven manner, leaving less room for hedonic engagement. Indeed, prior research on chatbots suggests that open-domain conversations, due to their less predictable and more exploratory nature, tend to evoke higher hedonic value and user enjoyment compared to structured, task-oriented dialogues (Følstad & Bjerkreim-Hanssen, 2023; Haugeland et al., 2022).

Beyond gratifications, use purpose also influenced perceptions of the chatbot's attributes. Participants who relied on chatbots primarily for information-seeking or conversational purposes perceived the chatbot to be more competent or expert than those using it for task execution. Engaging in exploratory Q&A or discussion likely exposes users to the chatbot's breadth of knowledge and reasoning ability, enhancing perceptions of expertise. Task-oriented users, by contrast, may interact in narrower domains or shorter exchanges. If the chatbot fails to complete a task, the failure is immediately apparent, thereby dampening perceived competence.

Similarly, perceived friendliness (warmth) was highest among users who engaged in conversational use, significantly more than those using chatbots for task-based assistance, with information-focused users falling in between. A likely explanation is that *conversational* interactions invite the chatbot to adopt a more informal and personal tone, often incorporating humor, empathy, or supportive language, leading users to perceived it as warmer and more humanlike. Task-based interactions, in contrast, tend to be transactional, making the chatbot appear more impersonal or mechanical. This pattern aligns with the CASA paradigm, which holds that when chatbots exhibit social cues, such as an informal tone, empathy, or humor, users respond as if interacting with a social actor, thereby attributing greater warmth and friendliness to the system (Nass & Moon, 2000).

Overall, these findings highlight that different usage purposes evoke distinct affordance-driven gratifications and perceptions of LLM chatbots. Conversational or exploratory uses amplify experiential gratifications such as *Immersive Realism* and *Fun*, fostering hedonic engagement. In contrast, goal- or information-oriented uses emphasize utilitarian gratifications such as *Adaptive Responsiveness*. These differentiated patterns suggest that chatbot design should be purpose-driven: systems aimed at information support should prioritize precision, coherence, and adaptive feedback, whereas conversational chatbots should focus on naturalness, empathy, and playful user experience to maximize enjoyment and perceived human-likeness.

#### 5.1.3. How affordance-driven gratifications shape outcomes

This study also demonstrates how specific affordance-based gratifications ultimately drive key user outcomes. The differential effects of these gratifications reveal nuanced dynamics in how individuals evaluate LLM-based chatbots. First, attitudes toward chatbots were primarily shaped by gratifications associated with modality, interactivity, and navigability affordances. Interestingly, agency-based gratifications, including *Personalized Filtering*, *Bandwagon*, and *Community-Building* did not significantly predict attitudes. This absence is theoretically noteworthy, given that agency is often identified as a central affordance in digital media (Sundar, 2008). While bandwagon and community-building are not core functionalities of LLM-chatbots, it is notable that *Personalized Filtering*, arguably a central function, also failed to influence attitudes. A plausible explanation is that users now see personalization as basic. As personalization and adaptive algorithms have become common across AI systems, users may treat such feature as a default, not a special, value-adding benefits.

In contrast, user satisfaction was solely driven by a single gratification: *Adaptive Responsiveness*, an interactivity-based gratification. Among all potential gratifications, only the chatbot's ability to adjust in real time, understand conversational context, and generate coherent contingent responses significantly predicted satisfaction. This suggests that user satisfaction depends less on the novelty or informational depth of content and more on the perception that the agent is attentively listening and adaptively responding. This aligns with research showing that responsiveness, alongside attentiveness, significantly enhances satisfaction in service encounters (De Ruyter & Wetzels, 2000), and by extension may operate similarly in chatbot interactions.

Perceived friendliness, however, drew upon four gratifications, each representing a different affordance type: *Immersive Realism* (modality), *Personalized Filtering* (agency), *Participatory Interaction* (interactivity), and *Fun* (navigability). On the other hand, perceived expertise closely mirrored the attitude pattern, relying on modality, interactivity, and navigability, particularly *Immersive Realism*, *Adaptive Responsiveness*, and *Browsing*. Notably, *Immersive Realism* (lifelike presence) contributes to both friendliness and expertise. Beyond this shared driver, friendliness and expertise are supported by different bundles of affordance-driven gratifications. Friendliness maps onto affective/relational cues, namely *Personalized Filtering* (individualized attention), *Participatory Interaction* (co-creation), and *Fun* (playful exploration). These gratifications make the agent feel socially present, attentive, and enjoyable, classic warmth cues. By contrast, expertise aligns with performance cues, specifically *Adaptive Responsiveness* (contingent, context-aware replies), and *Browsing* (breadth and navigability of information). These gratifications signal efficiency and informational reach.

Last, it is noteworthy that behavioral intention was driven by *Adaptive Responsiveness* (Interactivity) and *Fun* and *Browsing* (Navigability). This pattern suggests that users' willingness to continue engaging with LLM chatbots is shaped by interactional processes that unfold during use. *Adaptive Responsiveness*, the chatbot's ability to infer user intent, tailor replies, and repair misunderstandings, emerges as the most influential gratification because it meets a core psychological expectation of reciprocal and contingent communication. In human-AI interaction, such contingent responsiveness has been shown to evoke social presence, which in turn foster stronger engagement in AI chatbots (Cao et al., 2025). Unlike earlier conversational agents or traditional media, LLM-based chatbots can maintain context across turns, generate coherent follow-up responses, and dynamically adapt to user corrections. These capabilities transform the interaction from a transactional exchange into a fluid, co-constructed dialogue, thereby enhancing sustained use. The influence of *Fun* and *Browsing* further highlights the role of exploratory navigability in sustaining engagement. LLM chatbots' ability to help users navigate vast amounts of information effortlessly, and even playfully, reduce cognitive effort in information search and make information-seeking feel playful rather than laborious, which in turn promote sustained use.

### *5.2. Limitations*

This study has several limitations. First, due to survey length constraints, we did not directly assess participants' broader technical proficiency with AI beyond their chatbot usage. More tech-savvy users may seek different gratifications or hold higher expectations. However, given that a LLM chatbot is a relatively user-friendly system requiring minimal technical expertise, we suspect that technological proficiency did not substantially skew our results. Still, future research should include such measures to test this possibility.

Second, although our sample size ($N = 249$) was adequate for our exploratory factor analyses to identify gratification dimensions. Given the relatively large number of items compared to our sample size, conducting a confirmatory factor analysis (CFA) would have risked overfitting or unstable estimates. We deemed EFA the most appropriate approach for our exploratory goals. Nevertheless, once more stable factor structures are established and larger samples become available, future research should employ CFA or related confirmatory techniques to validate these gratification dimensions.

Third, we used a self-categorization approach to classify primary chatbot use, consistent with user-centered U&G logic. While this captures perceived functions, it also introduced subjectivity and potential inconsistencies, especially for multifunctional LLM systems. Future studies could combine self-reports with behavioral or log data to provide a more objective classification of chatbot use.

## 6. Conclusion

Despite these limitations, this study advances our understanding of how users engage with LLM chatbots by identifying distinct affordance-based gratifications and demonstrating their influence on important outcomes. Theoretically, the findings extend U&G research by showing that user motivations for chatbot use go beyond broad, traditional gratifications (like generic “entertainment” or “information”). Instead, users pursue specific gratifications enabled by LLM chatbot capabilities, such as immersion, adaptiveness, and participatory interaction, that reflect the unique affordances of this technology. This suggests that U&G frameworks must incorporate affordance-level mechanisms to account for how LLM chatbots shape motivations. In addition, the results illustrate that gratifications are *contextually activated* by users' primary purpose of use, reinforcing a central U&G principle while showing how distinct affordances (e.g., interactivity, navigability) become salient in different contexts. Finally, the strong effects of adaptive responsiveness highlight a new mechanism that has not been central in earlier U&G research but appears foundational in AI-mediated communication. Together, these insights refine and extend U&G theory by specifying how emerging AI affordances give rise to new motivational patterns and user experiences.

Practically, these insights suggest that designers should align chatbot affordances with users' primary purposes and desired outcomes. For information- or task-oriented use, prioritizing *Adaptive Responsiveness* (timely, context-aware feedback and coherent follow-ups) is most likely to enhance perceived expertise and satisfaction, thereby improving overall attitudes toward the system, and intentions for continued use. For conversational or exploratory use, emphasizing *Immersive Realism*, *Participatory Interaction*, and *Fun* (natural, personable dialogue; co-creation; playful exploration) can strengthen perceived friendliness (warmth) and hedonic engagement, which also foster favorable attitudes. In practice, designers should segment use cases, set explicit outcome targets, and tune affordances accordingly.

## CRediT authorship contribution statement

**Eun Go:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Taeyoung Kim:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation.

## Ethical approval and informed consent

This study involving human participants received approval from the Institutional Review Board (IRB) at Western Illinois University prior to data collection (approval number 032-25). Informed consent was obtained from all participants before their participation in the study.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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